Fiscal Year:	FY 2010	**Task Last Updated:**	FY 03/31/2010
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	NSCOR: Progressive Alterations of Central Nervous System Structure and Function Are Caused by Charged Particle Radiation		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	grnelson@llu.edu	**Fax:**	FY 909-558-4035
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8364
Organization Name:	Loma Linda University		
PI Address 1:	Basic Sciences, Div. Radiation Research 11175 Campus Street		
PI Address 2:	Chan Shun Pavilion, Room A-1024		
PI Web Page:			
City:	Loma Linda	**State:**	CA
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Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
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Key Personnel Changes/Previous PI:	Weili Lin, Ph.D. University of North Carolina. Subcontract terminated January 23, 2007. Micheal Pecaut, Ph.D. Loma Linda University. Resigned from project. 2007. Marco Stampanoni, Ph.D. Paul Scherrer Institute, Villigen, Switzerland. New subcontract Lead Investigator for X-ray Tomography. New Subcontract November, 2006.		
COI Name (Institution):	Chang, Polly Y. (SRI International) Fike, John R. (University of California San Francisco) Mao, Xao-Wen (Loma Linda University) Obenaus, Andre (Loma Linda University) Song, Sheng-Kwei (Washington University School of Medicine) Spigelman, Igor (University of California at Los Angeles) Stampanoni, Marco Ph.D. (Paul Scherrer Institute) Favre, Cecile (Loma Linda University) Vlkolinsky, Roman (Loma Linda University)		

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Task Description:	The experimental program detailed above will provide a comprehensive, quantitative assessment of the effects of charged particle radiation on the mammalian brain using the mouse hippocampus as a model. The mouse brain will be irradiated with accelerated protons and iron ions and examined over the course of two years for progressive changes in hippocampal structure and function. Cell populations in tissue sections will be quantified by a high precision stereology method to measure rates of loss of neurons, vascular endothelial cells and glia. Brain blood vessel topology will be further characterized by an erosion casting technique that produces a 3-D map of the vessels at 1 micron resolution. Stem cells and ongoing neurogenesis will be quantified using a complementary technique in which dividing cells are directly detected. Activation of cells and residual genetic damage in these cells will be characterized by immunofluorescence and mRNA level measurements of selected cytokine biomarkers and analysis of mutations in the transgene b-galactosidase. Functional impairments of neuronal ensembles will be quantified by electrophysiological measurements of hippocampal tissue slices and patch clamp measurements will detail changes in membrane properties. System level changes will be quantified by electroencephalography and two magnetic resonance imaging modalities that map white matter tracts (diffusion tensor imaging) and local metabolic rate (oxygen extraction). The plasticity of the brain as a system will be tested in its reaction to an immune system challenge mediated by lipopolysacharide. Finally the ability of radiation to modulate the latency of a neurodegeneration process will be tested in a transgenic mouse that overproduced amyloid precursor protein and develops an Alzheimer disease-like pathology. Taken together these results should provide a basis for determining risk to the CNS from charged particle radiation exposure. The main hypothesis for the grant is: Charged particle radiation exposure induces progressive deleterious structural and functional changes in the brain, leading to performance decline. Research Aims: Hypothesis (A) Charged particle irradiation causes a progressive loss of cells and a remodeling of CNS tissue as a function of dose, time and LET. To test hypothesis A we will: Specific Aim A1. Quantify the cellular composition of mouse brain vascular endothelium using stereological analysis of preserved tissue. Specific Aim A2. Quantify the composition of mouse neurons, selected glia and stem cells using stereological analysis of preserved tissue. Specific Aim A3. Quantify the activation of resident and infiltrating mouse immune cells using stereological analysis immune-labeled tissue. Specific Aim A4. Quantify key molecular biomarkers associated with radiation-induced damage to mouse cells and their microenvironment using targeted gene expression profiling and mutation analysis of a structural gene. Hypothesis (B) Charged particle irradiation alters the functional output or performance of the brain as a function of dose, time and LET. To test hypothesis B we will: Specific Aim B1. Quantify tissue-level electrophysiological properties of mouse brain using tissue slices that represent intact regional ensembles of interacting cells. (extracellular, patch clamp) Specific Aim B2. Map the time evolution of macroscopic structural features and metabolic alteration in mouse brains using MRI. Hypothesis (C) Exposure to radiation decreases disease latency and alters response to acute stressors. To test hypothesis C we will: Specific Aim C1. Quantify the magnitude and time course of electrophysiological changes in brain slices of mice expressing a human transgene that produces an Alzheimer disease-like pathology and limits output. Specific Aim C2. Quantify the electrophysiological performance of brain slices in mice reacting to the application of systemic immune stress using lipopolysaccharide as a surrogate infectious agent Foremost in the Critical Path Roadmap system for radiation effects on the central nervous system is Risk #29 – Acute and Late CNS Risks. All of the research and technology questions associated with this risk have priority level one for Mars missions and priority level 3 levels for ISS and 30d lunar missions. The questions addressed by the NSCOR project are 29a (functional changes), 29b (late degeneration), 29c (Alzheimer predisposition), 29e (late degeneration), 29g (cell and tissue mechanisms of damage), 29h (different cell populations including stem cells) and 29i (biomarkers). Also, a NASA workshop on CNS radiobiology identified supporting questions to assist in designing experiments to elucidate CNS effects in response to radiation exposure. These questions include: 1) What is the latency to CNS injury, 2) What cell types are involved in CNS injury (neurons, stem cells, glia)? and 3) What vascular changes occur in response to exposure. We have used this framework to focus our investigation on the effects of charged particles on the CNS. Specifically we will quantify cellular and functional changes in the hippocampus following high energy protons (main SPE constituent) and Fe-56 particles . We will use male rodents with sufficient statistical samples to address inter-individual variation and will address workshop questions 1-3 by examining the responses of different constituent cell types and the latency of damage. NSCOR subproject A quantifies different cell populations (29b, 29e, 29g, 29h); subproject B addresses functional changes and non-invasive techniques for identifying biomarkers (29a, 29i); and subproject C targets late degenerative changes with an Alzheimer model and the role of inflammation (29b, 29c, 29e, 29g and 29i). Our strategy is therefore aligned with the programmatic goals set forth by NASA and its advisory groups.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The response of the central nervous system to ionizing radiation will have direct impact on the development of radiotherapy using charged particles. The ability of protons, helium ions and carbon ions to be localized precisely in treatment fields close to critical structures (eg. optic nerve) make charged particle radiotherapy highly desireable. The limiting biological constraint on such treatments is the reaction of normal tissue surrounding a tumor site. The results of the NSCOR task will quantify cell loss and tissue functional changes form proton and iron irradiation that can be used directly to estimate normal tissue complication probabilities in the context of radiotherapy.
	Summary Results by Aim Aim A1. Significant time and dose dependent progressive loss of endothelial cells are seen along with loss and remodeling of microvessels; hippocampal subfields are differentially affected. This predicts that vascular insufficiency is occurring that may be manifest as poor perfusion and hypoxia with reduced performance of parenchymal cells. Observations for iron ions are recapitulated for animals exposed to protons. However, maximum cell loss is observed after 1 Gray rather than 0.5 Gray. A collaborative effort with University of Arizona Professor Timothy Secomb is allowing us to predict local tissue perfusion using a mathematical model developed by Dr. Secomb. Aim A2. Significant time and dose dependent loss of neural precursor cells is observed in a lineage-specific pattern. Loss of neural precursor cells is associated with cognitive impairment and spatial memory problems.

Task Progress:	Aim A3. Total microglial cell numbers are relatively unaffected by irradiation but sharp increases in newly-born microglia occur post-irradiation. Different phenotypic changes are seen in cells which are perivascular versus those that are not. This suggests that microglia are responding to persistent radiation-induced inflammatory changes or alterations to the microenvironment. Aim A4. Assessment of DNA damage in the hippocampi of transgenic mice shows that mutations continue to accumulate with time after irradiation indicating the continued presence of latent DNA damage. Their structural spectrum is LET and time dependent and differs from whole brain indicating regional specialization in adaptation to genotoxicity Gene expression profiling demonstrates activation of neurotrophic and adhesion factors associated with synaptic plasticity as well as up-regulation of chemokine receptors associated with inflammatory processes. Aims A1-4. Methods Summary: Perfusion Fixation/Euthanasia, Immunohistochemistry, Confocal Image acquisition, Gene expression studies, Mutation Analysis. Aim B1. Extracellular recordings from the CA1 region evaluated pyramidal cells after stimulation of CA3 axons. Input-output tests demonstrated time and dose dependent modifications to both dendritic and somatic region properties that are associated with enhanced excitability and decreased synaptic efficacy. Long-term potentiation protocols demonstrated a reduction of synaptic plasticity following irradiation that is consistent with a potential reduction of memory forming ability. In vitro patch clamp recordings showed that intrinsic membrane properties of neurons are resistant to radiation. Pharmacological manipulations then demonstrated changes in the inhibitory synapses on CA1 pyramidal cells suggesting that their hyperexcitability may be due in part to a reduction in GABAergic inhibition. Dysregulation of inhibition could be associated with seizure risk. Aim B2. Diffusion Tensor Imaging visualized and quantified altered structural brain changes based on altered water diffusion properties (direction, magnitude and anisotropy). Alterations in the corpus callosum, exterior capsule and hippocampus were observed and histological comparisons are underway to determine whether early aspects of demyelination or gliosis are being detected. Aims B1-3. Methods Summary: In Vitro Brain Slice Preparation, Extracellular Recordings and Long Term Potentiation (LTP), Patch Clamp Recordings, Histological Assessment, Diffusion Tensor Imaging. Aim C1. APP23 transgenic mice recreate most of the pathological features of Alzheimer's Disease, and most importantly, the disease progression is continuous over several months. These mice also feature prominent alterations of the vasculature, presumably one of the most vulnerable tissues in the brain after radiation. Electrophysiological measurements are showing that disease-related decreases in neuronal output and synaptic efficacy occur earlier in irradiated animals, which confirms our original hypothesis. This is consistent with the idea that that radiation may accelerate late neurogenerative changes . Using 3D vascular polymer cast technology combined with synchrotron based tomographic microscopy, vascular topology changes after irradiation have been detected. For the hippocampus, these changes are consistent with a change in vessel segment length and decreased spacing between vessels. As in Aim A1 this may be indicative of reduced perfusion and function of parenchymal cells. We expect to observe neovascularization and infarcts in older APP mice and will compare the time course of this progression in control and irradiated animals to further test whether radiation causes a decreased latency in disease progression. Amyloid Beta plaque load is being assessed using ThioS staining and preliminary results indicate an acceleration of plaque formation following irradiation. Aim C2. In order to assess the ability of the brain to respond to external environmental shocks and restore orderly normal function (homeostasis) we apply a controlled septic shock to cause an inflammatory reaction mediated by the periphery--nerves and immune system. The CNS response to peripheral lipopolysaccharide (LPS) exposure has long been established and typically includes changes in electrophysiology and cytokine expression. We found that in irradiated animals, the patterns of electrophysiological changes associated with reactions to LPS are complex and unlike those of either LPS or irradiation alone. We had expected the effects to be additive or synergistic. This indicates that radiation causes plastic changes to the CNS that persist for many months in the mouse and alter its program of response to septic shock. This is consistent with the idea that irradiation may potentiate the risks from late secondary insults which might include trauma or infection. Aims C1-C2. Methods Summary: In vitro electrophysiology as in Aim B1, Extracellular Protocols and Analysis of Neuronal Activity, Corrosion casting, Synchrotron X-ray microCT (XTM), Confocal microscopy, Image analysis, LPS Administration. Complementary experiments assessing mechanisms and establishing baseline for new NSCOR project. Administration of fluorescent hypoxia marker piminidazolento irradiated animals demonstrated dose and time dependent hypoxic status in hippocampus in regional patterns matching microvessel sensitivity. This supports the idea that at higher doses a hypoxic episode evokes neoangiogenesis. A microelectrode array system (Panasonic MED64) has been installed and used successfully to simultaneously record from multiple sites in the hippocampus circuitry allowing characterization of network-level activity. A hypoxia reperfusion protocol with irradiated hippocampal slices has shown an interaction between tissue responses to oxidative stress and radiation. Electrophysiological experiments with geriatric irradiated rats shows age-related sensitivity to charged particles for electrophysiology parameters including long term potentiation. Rat electrophysiology responses parallel those in mice. Eyes of geriatric irradiated rats were analyzed for late degeneration using diffusion tensor imaging and exhibited retinal and optic nerve changes.
Bibliography Type:	Description: (Last Updated: 03/26/2026)
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